Stock Tracker VM — Migration from Default VPC to Custom VPC

Project: `my-generic-project-504019` **Date:** 2026-08-14 **VM:** `stockapp-vm` (zone: `us-central1-a`, machine type: `e2-micro`, free tier) **Result:** ✅ Successful, zero data loss, same external IP retained

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Why

Moved `stockapp-vm` off the auto-created `default` VPC and into a purpose-built custom VPC (`stockapp-vpc`), as a hands-on exercise in VPC design, firewall scoping, and safe VM lifecycle management. GCP does not support live-migrating a VM's network interface between VPCs — the VM must be recreated, with its boot disk preserved and reattached.

What Changed

Item	Before	After
VPC	`default`	`stockapp-vpc`
Subnet	`default` (auto, `10.128.0.0/20`)	`stockapp-subnet` (`10.10.0.0/24`)
Internal IP	`10.128.0.2`	`10.10.0.2`
External IP	`136.65.201.133` (ephemeral)	`136.65.201.133` (now static, reserved)
SSH access	Open to `0.0.0.0/0`	Restricted to a single `/32` (your IP)
RDP rule	Present network-wide (`default-allow-rdp`)	Not recreated — not needed for this Linux VM
Boot disk	`stockapp-vm` disk, `autoDelete=False`	Same disk, reattached, `autoDelete=False` preserved
App data (SQLite DB, PDFs)	On boot disk	Untouched — carried over on the same disk

Steps Performed

1. **Reserved the existing external IP as static** `gcloud compute addresses create stockapp-static-ip --addresses=136.65.201.133 --region=us-central1`
2. **Created custom VPC + subnet** `stockapp-vpc` (custom mode), `stockapp-subnet` (`10.10.0.0/24`, `us-central1`)
3. **Created firewall rules on `stockapp-vpc`**, matching prior exposure (SSH tightened):
    - `stockapp-vpc-allow-80` — TCP 80, `0.0.0.0/0`, target tag `stockapp`
    - `stockapp-allow-ssh` — TCP 22, restricted to home IP `/32`, target tag `stockapp`
    - `stockapp-allow-icmp` — ICMP, target tag `stockapp`
    - `stockapp-allow-internal` — all TCP/UDP/ICMP within `10.10.0.0/24`
4. **Verified boot disk `autoDelete` was `False`** before touching anything (it was)

5. **Stopped and deleted the old VM**, disk survived (confirmed via `gcloud compute disks list`)
6. **Recreated the VM** in `stockapp-vpc` / `stockapp-subnet`, reattaching the original disk as boot disk and the reserved static IP:

```
gcloud compute instances create stockapp-vm \
  --zone=us-central1-a --machine-type=e2-micro \
  --network=stockapp-vpc --subnet=stockapp-subnet \
  --tags=stockapp \
  --disk=name=stockapp-vm,boot=yes,auto-delete=no \
  --address=stockapp-static-ip
```

7. **Verified**: app loads at `http://136.65.201.133`, SSH connects from home IP, network field confirmed as `stockapp-vpc`.

Cleanup (optional, not yet done)

Old orphaned rule on `default` VPC, no longer used since the VM moved:

```
gcloud compute firewall-rules delete allow-stockapp-80 --project=my-generic-project-504019
```

Left in place (network-wide, may serve other resources like the `gke-dinosite-cluster` workload in this project): `default-allow-ssh`, `default-allow-rdp`, `default-allow-icmp`, `default-allow-internal`

Notes for Next Time

- Firewall rule names must be unique **per project**, not per VPC — collided once with `allow-stockapp-80`, renamed to `stockapp-vpc-allow-80`.
- Always check `disks[0].autoDelete` before deleting a VM you want to keep the disk from.
- Reserving the external IP *before* deleting the old VM is what let the same IP carry over.